L11: Knowledge Check

1 Multiple choice 1 point

For a given fraction of randomly removed nodes f, which of the following networks has the **second-largest** connected component?

- Random G(n,p) network
- Power-law network
- Ring network
- Clique network

2 Multiple choice 1 point

According to the Molloy-Reed criterion, which of the following conditions should be true so that we have a giant component in a random network that follows the configuration model?

- The average degree of a node should be at least 2
- The average degree of a node in the largest connected component should be at least 2
- The average degree of a node in the largest connected component should be at least 1
- The average neighbor degree should be at least 1

3 Multiple choice 1 point

In the decentralized search problem, if we have a 3-dimensional grid, what is the optimal value of the exponent q?

- 2
- 3
- 6
- 8

4 Multiple choice 1 point

Which of the following statements are **TRUE**?

- The network in Kleinberg's model is the same with the Watts-Strogatz model in Lesson-5
- In the two-dimensional Kleinberg model, a network generated with a large value of the exponent q will not have small world properties
- In Kleinberg's model, if q=0, the random edges connect nodes in a manner that is dependent on the distance between them
- In Kleinberg's model, a lower q will be better in terms of expected delivery time

5 Multiple choice 1 point

In the Kuramoto model, which term directly quantifies how coherent the various oscillators are?

- The complex number $R(t)e^{i\Phi(t)}$
- The phase $\Phi(t)$
- The order parameter $R(t)$
- The average phase of the oscillators

6 Multiple choice 1 point

Which of the following statements are **TRUE**?

- For a power-law network with degree exponent γ, the network will always be synchronized if $\gamma > 3$
- In the Kuramoto model with two oscillators, they cannot synchronize if their angular frequency ω differs by more than K
- In the Kuramoto model, as we increase the coupling strength K, the system moves from a random state to a coherent state gradually
- It is easier to synchronize a regular grid network of oscillators than a power-law network

7 Multiple choice 1 point

Suppose we are given a power-law network with $k_{min} = 4$ and degree exponent 4. Which of the following is correct about the attack critical threshold f_c ?

- 0.3 < f_c < 0.65
- 0.1 < f_c < 0.45
- 0.25 < f_c < 0.5
- 0.65 < f_c < 0.9